

Md. Mushfiqur Rahman

Environmental Systems Engineer (MAsc Candidate) • Sustainable Materials • Water & Data

St. John's, NL, Canada
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Portfolio • LinkedIn • ResearchGate

ABOUT ME

My path into environmental engineering started on construction sites in Bangladesh. As a Building Engineering & Construction Management student at KUET, I researched how to turn industrial and demolition waste into high-strength, low-carbon concrete, work that led to five publications. As an engineer in Dhaka I inspected sites, wrote environmental compliance reports and put energy-efficiency measures into practice. I then moved to Canada to study how the built environment and natural systems interact. At Memorial University I now apply machine learning to water quality, building models that predict how wildfires affect the rivers communities drink from. I bring field experience, materials research and data skills together, and I'm looking for a Winter 2027 co-op in environmental management, water, construction or infrastructure.

EDUCATION

Master of Applied Science (Co-op), Environmental Systems Engineering & Management 2025 – 2027 (expected)
Memorial University of Newfoundland, St. John's, NL CGPA 3.87
Coursework: Environmental Risk Assessment • Environmental Systems Engineering • Environmental Management Systems (ISO 14001) • Environmental Statistics • Environmental Laboratory • Pollution & Mitigation • Environmental Impacts of Offshore Oil & Gas • Environmental Chemistry & Toxicology

B.Sc. in Building Engineering & Construction Management 2019 – 2024
Khulna University of Engineering & Technology (KUET), Bangladesh CGPA 3.33/4.00 • WES certified
Major: Structural & Materials Engineering • University Merit Scholarship & Stipend (2019–2022)

RESEARCH & TECHNICAL PROJECTS

Quantifying Wildfire Impacts on Source-Water Quality: A Machine-Learning Framework Jan 2026 – Present
Memorial University, ENGI 960A/B Engineering Project • Supervisor: Dr. Arpana Rani Datta • team of four

- Built a turbidity-prediction framework on 823 water-quality samples (4 stations, 2000–2026, Peace–Athabasca Basin) with 22 chemistry, time and station predictors.
- Compared Random Forest, Extra Trees, Gradient Boosting and XGBoost; Extra Trees performed best ($R^2 = 0.775$). Feature importance identified nitrogen, colour and metals (Mn, Al, Fe) as the dominant drivers, consistent with post-fire runoff.
- Tested transfer to Labrador's Churchill River: the Alberta-trained model over-predicted about 17×, showing that the local hydrology–water-quality link must be established before a model can be reused in a new watershed.

Undergraduate Thesis: High-Strength Concrete with Recycled Aggregate & Metakaolin 2023 – 2024

- Designed and tested ten concrete mixes; the optimal blend reached 60.36 MPa while replacing ~45% of natural aggregate with demolition waste, cutting embodied carbon.

Design Projects, KUET 2022 – 2024

- BIM design and energy modelling of a 4-storey building (Revit, Green Building Studio, Robot Structural Analysis, MS Project); structural analysis of a 12-storey tower under wind and seismic loads (ETABS).
- Technical literature review of 3D-printed (additively manufactured) concrete: materials, printing methods, performance and applications (2022).

PROFESSIONAL EXPERIENCE

- Assistant Engineer

Jun 2024 – Feb 2025

ZAS Corporation Ltd, Dhaka, Bangladesh

 - Authored 20+ environmental compliance reports and technical submissions with a 100% first-pass approval rate on safety and permitting reviews.
 - Inspected fire-safety systems across multiple sites, closing 30+ code non-conformities with a zero-incident safety record.
 - Designed energy-efficiency and renewable-integration measures to reduce building energy use and operating costs.
 - Coordinated cross-functional teams and tracked milestones in MS Project, delivering 95%+ of milestones on schedule.
- Engineering Intern

May 2024

Sheltech Pvt. Ltd, Dhaka, Bangladesh

 - Carried out site inspections and quality-control checks that flagged defects early and kept work aligned with safety and environmental protocols.
 - Produced 30+ AutoCAD drawings and technical documentation packages for site crews.

PUBLICATIONS (5 · 77 citations on ResearchGate, Sept 2026)

- Sobuz, M. H. R., **Rahman, M. M.**, et al. (2025). Combined influence of modified recycled concrete aggregate and metakaolin on high-strength concrete production: Experimental assessment and machine learning quantifications with advanced SHAP and PDP analyses. *Construction and Building Materials*, 461, 139897. doi:10.1016/j.conbuildmat.2025.139897
- Rahman, S., Rana, M. J., **Rahman, M. M.**, et al. (2025). Investigating the energy-efficiency of Glauber's salt as PCM-enhanced mortar in building envelope. *Advances in Building Energy Research*, 19(6), 798–825. doi:10.1080/17512549.2025.2547399
- **Rahman, M. M.**, & Hossain, M. S. (2024). Fresh and hardened properties of high-strength concrete using coal bottom ash and silica fume. Conference paper.
- **Rahman, M. M.**, & Hossain, M. S. (2024). Enhancing concrete mechanical performance through metakaolin and improved recycled concrete aggregates. Conference paper.
- Ashraf, J., **Rahman, M. M.**, & Rana, M. J. (2023). Impact of exterior facade materials on energy performance of residential structures in Bangladesh. *Proc. ICPACER 2023*.

TECHNICAL SKILLS

Environmental	Environmental risk assessment · compliance & HSEQ · EMS / ISO 14001 · life-cycle & carbon-footprint reduction · water-quality analysis · energy auditing · regulatory & technical writing
Data & ML	Python (pandas, scikit-learn, XGBoost, seaborn) · Random Forest, Extra Trees, Gradient Boosting, k-NN · statistical analysis · Origin Pro · Excel · C · LaTeX
GIS	ArcGIS (currently building skills and working toward certification)
Engineering & BIM	Autodesk Revit · Green Building Studio · AutoCAD · ETABS · Robot Structural Analysis · Navisworks · SketchUp · Lumion · materials testing
Project tools	MS Project · Primavera P6 · Microsoft 365 (Word, Excel, PowerPoint, Teams, SharePoint) · Adobe Photoshop, Illustrator, Premiere Pro

CERTIFICATIONS, AWARDS & LANGUAGES

Certifications: Machine Learning Specialization, Stanford University (2024) · Construction Project Management, Columbia University (2024) · Standard First Aid	Awards & service: University Merit Scholarship & Stipend, KUET (2019–2022) · Student Volunteer, 20+ hours, Bronze recognition, Memorial University (2025–present)
Languages: English (IELTS 7.5, CEFR C1) · Bengali (native)	Work status: Based in St. John's, NL · eligible to work in Canada · available for Winter 2027 co-op

References available on request.